Internet Appendix

What moves stock prices?
The role of news, noise, and information

October 2021

Section 1 of this Internet Appendix reports additional tests of the relation between information and noise. Section 2 reports the variance components using different types of weighted averages when aggregating across stocks. Section 3 reports results for variance levels, rather than variance shares. Section 4 examines the optimal lag length in the VAR and tests the robustness of the results to alternative lag specifications. Section 5 examines the robustness of the main results to an alternative measure of signed trading volume. Section 6 tests how dealer collusion on NASDAQ during the 1990s impacted noise in prices. Section 7 examines the robustness of the main results to winsorizing the variance components at a lower level. Section 8 contains the theoretical model (and proofs) that motivate the decomposition of variance into several types of information and noise. Section 9 derives the effects of alternative assumptions about the covariance between information and noise.

1. Relation between information and noise

Theory predicts that information and noise are related. Noise is needed for markets to be liquid (e.g., Kyle, 1985; Glosten and Milgrom, 1985) and for private information to show up in prices (e.g., Grossman and Stiglitz, 1980; Kyle 1989). In the words of Black (1986): "Noise makes financial markets possible, but also makes them imperfect."

The relation between information and noise in theory is symbiotic: an exogenous increase in noise trading tends to make markets more liquid, thereby incentivizing more information acquisition (e.g., Grossman and Stiglitz, 1980; Kyle 1984, 1989; Admati and Plfeiderer, 1988). Going the other way, an exogenous increase in informed trading that tends to bring more private information into prices at a faster rate, imposes adverse selection risk on liquidity providers, causing a deterioration in liquidity and potentially more noise in prices (e.g., Kyle, 1985; Glosten and Milgrom, 1985).

Therefore, based on theory, we expect a positive relation between the variance attributable to information and the variance attributable to noise. In contrast, variance shares will, by construction, have a negative relation between the share of information and the share of noise as one is the inverse of the other. Therefore, in the following empirical analysis, we test the relations between the different variance components measured in levels.

The most basic test is a simple correlation in the full sample (1960-2015, all US listed stocks). The correlation between the noise and information components of variance in the pooled stock-year panel is 0.74, indicating a strong positive relation between information and noise, consistent with theory.

Next, we regress the noise component of variance on the estimated information in prices using various specifications so that we can examine the relation in more detail. We examine the relation with each of the different types of information (market-wide information, private firm-specific information, and public firm-specific information) and the total information in prices, being the sum of the three information components. The results are in Table A.1.

In Panel A we control for stock fixed effects to examine whether changes in the information content through time are related to changes in the amount of noise through time. The results show a positive and significant relation between noise and total information, as well as each type of information. In Panel B we control for year fixed effects to focus on the cross-sectional variation in noise and information. There too we find a significant positive relation between noise and information, consistent with theory.

Finally, in Panel C, we control for a wide range of stock and time-series variables that we use in the paper (Table 3) including a post-1997 indicator variable, the stock price, size, and industry, two existing measures of informational efficiency (overnight/intraday variance ratio and the delay metric), institutional holdings, hard to value stocks, two measures of liquidity (Amihud's (2002) ILLIQ and effective bid-ask spreads) and leverage. The positive and significant relation between noise and information remains robust to controlling for these variables.

Table A.1. Relation between information and noise.

This table reports the results from panel regressions of stock-year observations in which the dependent variable is the stock return variance attributable to noise ($Noise_{i,t}$). Models 1-4 examine how the noise in returns is related to the total amount of information in prices ($Info_{i,t}$; Model 1) and different types of information including market-wide information ($MktInfo_{i,t}$; Model 2), private firm-specific information ($PrivateInfo_{i,t}$; Model 3), and public firm-specific information ($PublicInfo_{i,t}$; Model 4). The components of variance are measured in terms of standard deviations of daily percentage returns. Panel A includes stock fixed effects. Panel B includes year fixed effects. Panel C includes several additional independent variables. Specifically, D_t^{POST} is a dummy variable that takes the value of one after 1997 and zero before. $lnP_{i,t}$ is the log price and $lnMC_{i,t}$ is the log market capitalization. $D_i^{Consumer}$, $D_i^{Healthcare}$, D_i^{HiTech}, and $D_i^{Manufact}$ are dummy variables that indicate the firm's industry group (the Other Industry grouping is the omitted category). $VarianceRatio_{i,t}$ is a measure of the amount of information impounded in prices during the trading session (volatility of open-to-close returns divided by the volatility of overnight close-to-open returns). $Delay_{i,t}$ is a measure of the delay with which a stock's prices respond to market-wide information. $InstoHold_{i,t}$ is the percentage of outstanding shares held by institutional investors. $HardToValue_{i,t}$ is a proxy for hard to value stocks, estimated as the first principal component of share turnover, idiosyncratic volatility, and firm age (following Kumar, 2009). $ILLIQ_{i,t}$ is the Amihud (2002) measure of illiquidity. $BidAskSpread_{i,t}$ is the average effective bid-ask spread. $Leverage_{i,t}$ is (long-term debt + current liabilities)/(long-term debt + current liabilities + shareholder's equity). The sample includes stocks listed on NYSE, AMEX, and NASDAQ from 1960 to 2015. T-statistics are in parentheses using standard errors clustered by stock and year. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels.

Panel A: Models with stock fixed effects				
	Model 1 $Noise_{i,t}$	Model 2 $Noise_{i,t}$	Model 3 $Noise_{i,t}$	Model 4 $Noise_{i,t}$
$Info_{i,t}$	0.51 (12.64)***			
$MktInfo_{i,t}$		0.59 (19.79)***		
$PrivateInfo_{i,t}$			0.47 (22.23)***	
$PublicInfo_{i,t}$				0.47 (25.1)***
R^2	45.6%	30.8%	15.0%	20.7%
Fixed Effects	Stock	Stock	Stock	Stock

Panel B: Models with time fixed effects				
	Model 1 $Noise_{i,t}$	Model 2 $Noise_{i,t}$	Model 3 $Noise_{i,t}$	Model 4 $Noise_{i,t}$
$Info_{i,t}$	0.52 (16.22)***			
$MktInfo_{i,t}$		0.61 (20.44)***		
$PrivateInfo_{i,t}$			0.58 (31.51)***	
$PublicInfo_{i,t}$				0.58 (35.85)***
R^2	58.5%	33.9%	37.1%	44.7%
Fixed Effects	Year	Year	Year	Year

Panel C: Models with an extended set of stock characteristics and control variables				
	Model 1	Model 2	Model 3	Model 4
	$Noise_{i,t}$	$Noise_{i,t}$	$Noise_{i,t}$	$Noise_{i,t}$
Intercept	1.06	1.46	1.71	1.32
	(4.85)***	(5.34)***	(6.40)***	(5.83)***
$Info_{i,t}$	0.20			
	(18.45)***			
$MktInfo_{i,t}$		0.49		
		(14.85)***		
$PrivateInfo_{i,t}$			0.31	
			(12.01)***	
$PublicInfo_{i,t}$				0.36
				(13.86)***
D_t^{POST}	-0.075	-0.033	0.029	-0.057
	(-1.15)	(-0.48)	(0.35)	(-0.83)
$lnP_{i,t}$	-0.13	-0.25	-0.23	-0.17
	(-3.45)***	(-5.40)***	(-4.32)***	(-4.07)***
$lnMC_{i,t}$	-0.041	-0.029	-0.043	-0.023
	(-3.78)***	(-2.12)**	(-3.25)***	(-1.95)*
$D_i^{Consumer}$	-0.029	0.071	0.01	0.016
	(-1.04)	(2.69)***	(0.29)	(0.50)
$D_i^{Healthcare}$	-0.11	0.055	-0.058	-0.041
	(-3.51)***	(1.77)*	(-1.45)	(-1.12)
D_i^{HiTech}	-0.096	0.03	0.008	0.009
	(-2.56)**	(0.78)	(0.14)	(0.19)
$D_i^{Manufact}$	-0.022	0.054	0.042	0.037
	(-0.77)	(1.90)*	(1.35)	(1.35)
$VarianceRatio_{i,t}$	-0.033	-0.016	-0.016	-0.026
	(-4.21)***	(-2.00)**	(-1.92)*	(-3.26)***
$Delay_{i,t}$	0.33	0.72	0.15	0.12
	(4.50)***	(7.14)***	(1.84)*	(1.76)*
$InstoHold_{i,t}$	-0.001	-0.002	-0.002	-0.002
	(-2.44)**	(-3.66)***	(-2.37)**	(-2.21)**
$HardToValue_{i,t}$	-0.003	0.11	0.10	0.07
	(-0.11)	(4.02)***	(3.07)***	(2.43)**
$ILLIQ_{i,t}$	0.003	0.004	0.004	0.003
	(3.36)***	(3.21)***	(3.39)***	(3.19)***
$BidAskSpread_{i,t}$	0.010	0.012	0.012	0.010
	(3.37)***	(3.39)***	(3.33)***	(3.27)***
$Leverage_{i,t}$	-0.001	0.00	0.00	-0.002
	(-0.24)	(0.00)	(0.14)	(-0.89)
R^2	64.39%	60.77%	56.52%	61.43%

2. Alternative weighting in market aggregates

In the baseline results, when we aggregate the variance shares to market-level averages we use variance-weighted averages consistent with Morck, Yeung, and Yu (2000, 2013). Here, we report results using equal-weighted averages as an alternative way to aggregate across stocks.

Table A.2 below is the equivalent of Table 2 in the paper but using equal weighting. The main difference between the two tables is that relative to the variance-weighted means in the paper, the equal-weighted average noise share is lower (23.35% vs 30.71%), the equal-weighted average private firm-specific information share is higher (27.75% vs 24.00%), and the equal-weighted average market-wide information share is also higher (13.16% vs 8.62%). These differences occur because noise shares tend to be higher for more volatile stocks and correspondingly, private information and market-wide information shares tend to be lower for such stocks. While the results suggest that different weighting schemes when aggregating across stocks produce different market-wide estimates, the main conclusions still hold: noise is a substantial driver of stock returns (between one-quarter and one-third of the variance depending on the weighting), market-wide information plays a relatively small role (8-13% overall depending on the weighting), and firm-specific information is the dominant driver.

The equal-weighted averages in Panels B, C, and D of Table A.2 show similar tendencies to those reported in the paper. The share of firm-specific information is higher in the post 1997 period and the noise share is lower in the post 1997 period. Small stocks tend to have noisier prices, and proportionally less market-wide information and private firm-specific information in prices. The differences in variance shares across industries are not large.

Next, we examine how the overall time-series trends change if we take equal-weighted averages. Figures A.1, A.2, and A.3 are equivalent to Figures 3, 5, and 6 in the paper but using equal weighting. The overall trends are the same, just the magnitudes are slightly different. The equal-weighted means show that noise peaks in the 1990s around the time of dealer collusion on NASDAQ and before tick sizes were reduced and has fallen substantially in the subsequent years. Firm-specific information in prices tends to increase through time, in particular the share of public information, consistent with increased disclosure and information availability. Market-wide information fluctuates through time and tends to spike during crises. Small stocks have noisier prices and the variance shares across industries are remarkably similar.

Table A.2. Stock return variance shares with equal weighting of stocks.
This table reports the mean variance shares (expressed as percentages of variance) for the period from 1960 to 2015. Stock return variance is decomposed into market-wide information ($MktInfoShare$), private firm-specific information ($PrivateInfoShare$), public firm-specific information ($PublicInfoShare$), and noise ($NoiseShare$). Panel A reports full sample averages. Panel B splits the sample into two sub-periods from 1960 to 1996, and from 1997 to 2015. Panel C groups stocks into quartiles by size (market capitalization) with quartiles formed separately each year. Panel D groups stocks into major industry groups: the *Consumer* group comprises the industries Consumer Durables, NonDurables, Wholesale, Retail, and some Services (Laundries, Repair Shops); the *Healthcare* group comprises the industries Healthcare, Medical Equipment, and Drugs; the *Manufact* group comprises the industries Manufacturing, Energy, and Utilities; the *HiTech* group comprises the industries Business Equipment, Telephone and Television Transmission; and the *Other* group comprises all other industries. The variance component shares are calculated separately for each stock in each year then averaged across stocks within the corresponding quartile or group, using an equal-weighted mean. We also report the differences in means for the post-1997 period minus the pre-1997 period (Panel B) and quartile 1 minus quartile 4 (Panel C) and corresponding t-statistics in parentheses. ***, **, and * indicate statistically significant differences at the 1%, 5%, and 10% levels using standard errors clustered by stock and by year. The sample consists of stocks listed on NYSE, AMEX, and NASDAQ.

	$MktInfoShare$ (%)	$PrivateInfoShare$ (%)	$PublicInfoShare$ (%)	$NoiseShare$ (%)
Panel A: Full sample				
	13.16	27.75	35.73	23.35
Panel B: Sub-periods				
1960-1996	13.03	27.09	34.70	25.19
1997-2015	13.36	28.70	37.23	20.72
Difference (Post-Pre 1997)	0.34	1.61	2.53	-4.48
	(0.19)	(1.27)	(2.55)**	(-3.52)***
Panel C: Quartiles by size (market capitalization)				
Q1=low	6.17	25.49	37.12	31.22
Q2	10.22	26.77	37.44	25.56
Q3	14.98	28.25	36.36	20.41
Q4=high	21.29	30.47	32.02	16.22
Difference (Q1-Q4)	-15.12	-4.98	5.10	15.00
	(-17.24)***	(-3.86)***	(5.25)***	(10.77)***
Panel D: Industry groups				
Consumer	13.12	28.39	35.75	22.74
Healthcare	11.10	31.18	37.00	20.71
HiTech	13.48	29.42	35.70	21.40
Manufact	15.32	28.34	34.86	21.48
Other	11.90	25.08	36.09	26.93

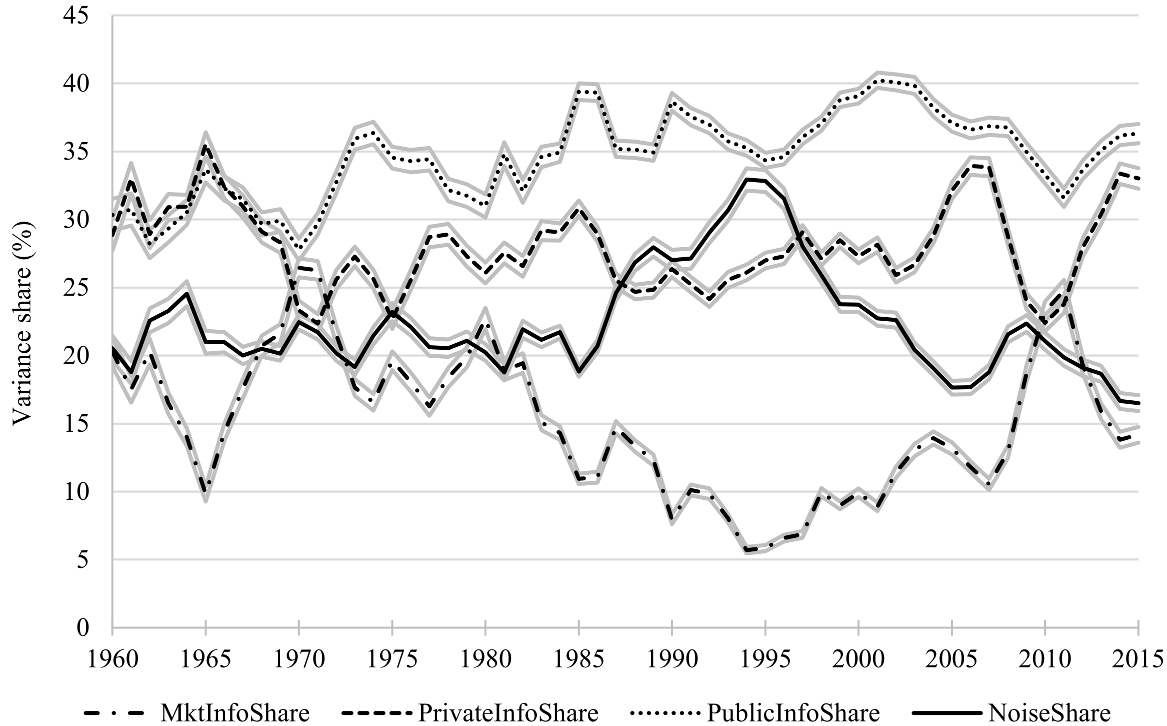


Figure A.1. Stock return variance components for US stocks through time using equal weighting.
This figure shows the time-series trends in the percentage of stock return variance that is attributable to noise ($NoiseShare$), market-wide information ($MktInfoShare$), trading on private firm-specific information ($PrivateInfoShare$), and public firm-specific information ($PublicInfoShare$). The variance shares are calculated separately for each stock in each year based on a VAR model. For each of the variance shares in each year we then take an equal-weighted average across stocks. Light gray lines provide 99% confidence intervals. The sample consists of stocks listed on NYSE, AMEX, and NASDAQ from 1960 to 2015.

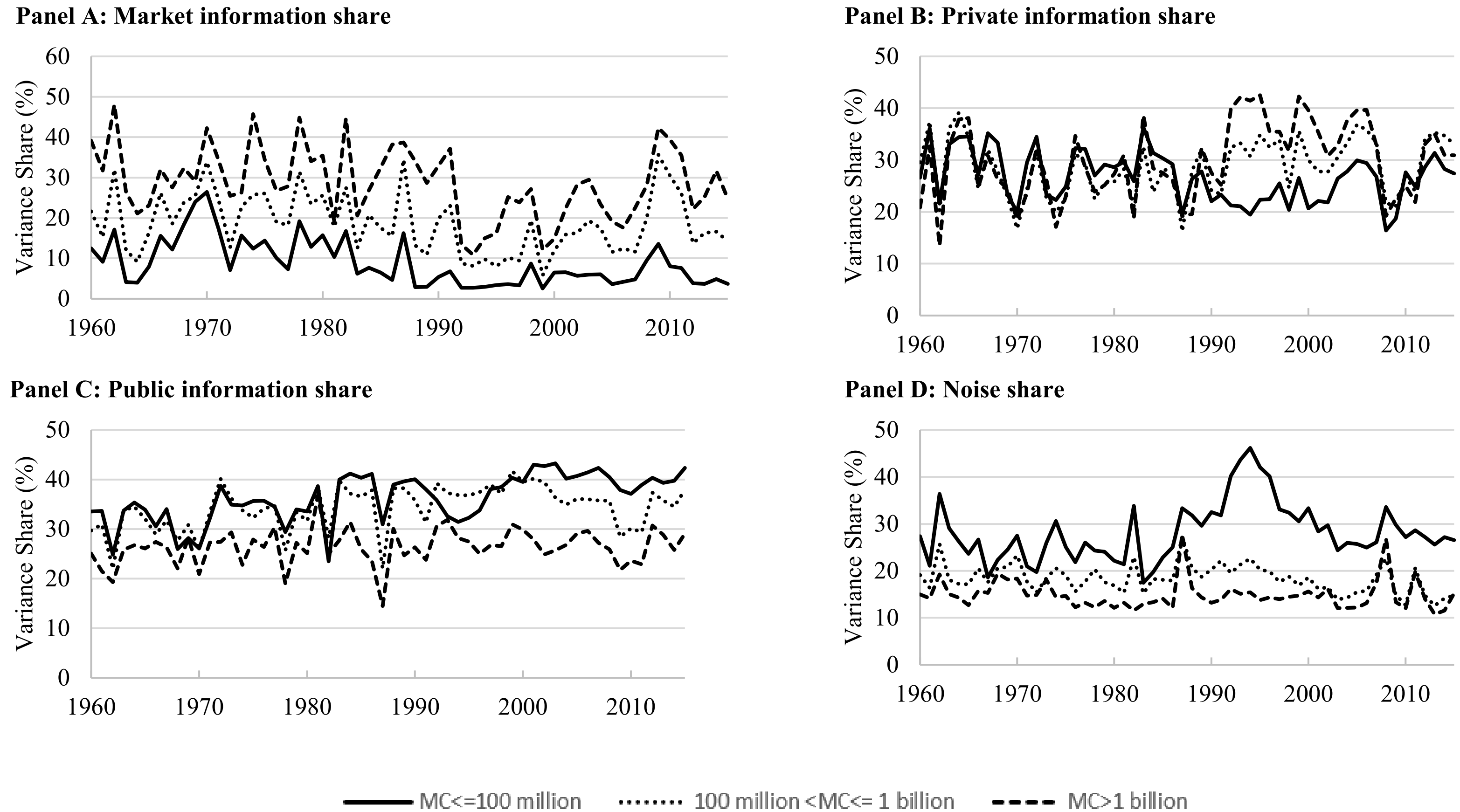


Figure A.2. Variance components in size groups through time using equal weighting.
This figure shows the time-series trends in the percentage of stock return variance that is attributable to market-wide information (Panel A), private firm-specific information (Panel B), public firm-specific information (Panel C), and noise (Panel D) in three market capitalization groups: stocks with market capitalization less than $100 million, market capitalization between $100 million and $1 billion, and market capitalization greater than $1 billion. These breakpoints are in 2010 dollars and are adjusted for inflation forward and backward in time using the GDP price deflator. Each year stocks are assigned to one of the three groups based on their market capitalization at the start of the year. The variance component shares are calculated separately for each stock in each year then averaged for each size group in each year, using an equal-weighted mean. The sample consists of stocks listed on NYSE, AMEX, and NASDAQ from 1960 to 2015.

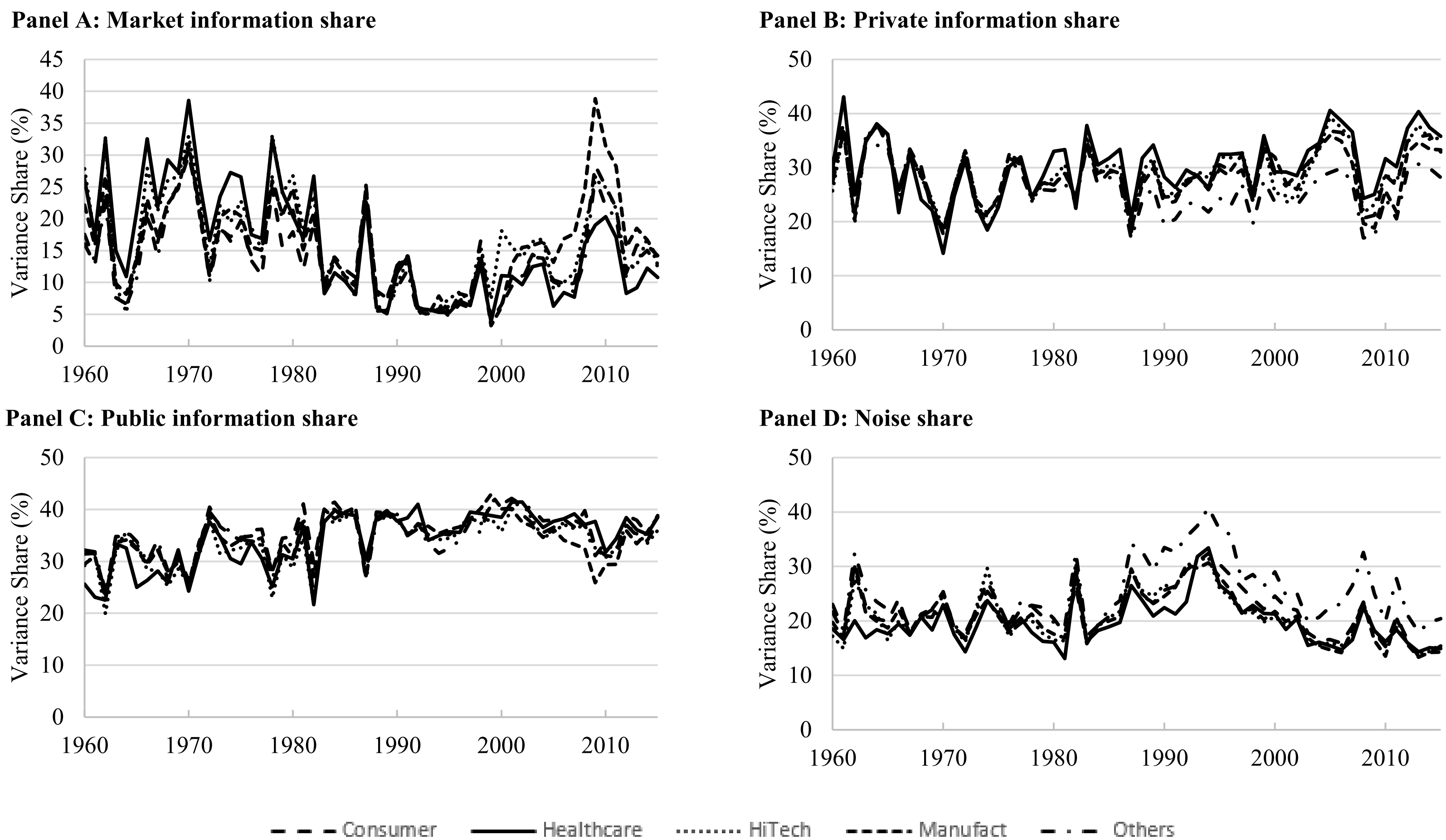


Figure A.3. Variance components in major industry groups through time using equal weighting.
This figure shows the time-series trends in the percentage of stock return variance that is attributable to market-wide information (Panel A), private firm-specific information (Panel B), public firm-specific information (Panel C), and noise (Panel D) in five major industry groups. The *Consumer* group comprises the industries Consumer Durables, NonDurables, Wholesale, Retail, and some Services (Laundries, Repair Shops); the *Healthcare* group comprises the industries Healthcare, Medical Equipment, and Drugs; the *Manufact* group comprises the industries Manufacturing, Energy, and Utilities; the *HiTech* group comprises the industries Business Equipment, Telephone and Television Transmission; and the *Other* group comprises all other industries. The variance component shares are calculated separately for each stock in each year then averaged for each industry group in each year, using an equal-weighted mean. The sample consists of stocks listed on NYSE, AMEX, and NASDAQ from 1960 to 2015.

3. Additional results for variance levels rather than shares

The variance decomposition produces estimates of the amount of stock return variance that is attributable to different types of information and noise. These estimates can be expressed in levels (variance) or as shares (levels normalized by total variance so that the shares sum to 100%). Throughout the paper, we report most results in terms of variance shares because they address the questions of "what moves stock prices?" or "what drives stock return variance?".

Here we report results for our main tables/figures using variance levels instead of shares.[1] Often the conclusions that we draw for variance shares also hold for variance levels. For example, if information and noise have equal levels, say information=2 and noise=2 giving an information share of 50% (2/4) and a noise share of 50% (2/4) and then the level of noise increases to say 4 then the noise share also increases, from 50% (2/4) to 67% (4/6).

However, there can be situations in which the variance shares and levels give different results. For example, in the previous illustration, suppose that when the level of noise increases from 2 to 4, information also increases but by a larger amount, say from 2 to 6. In this scenario, the noise share will fall from 50% (2/4) to 40% (4/10) even though there was an increase in the level of noise.

The following subsections therefore report our key tables/figures using variance levels.

3.1. Overall variance levels by firm size and industry

Table A.3 is the analogue of Table 2 in the paper and shows the overall means of the variance levels, including a breakdown by size and industry. All three variance components associated with information (market-wide information, private firm-specific information, and public firm-specific information) are higher in the post 1997 period and the differences are statistically significant. This finding is consistent with the observations made in the paper using variance shares and supports the notion that stock prices have become more informative through time. The average level of noise variance is lower in

[1] We convert the variance components into standard deviations (taking their square root) and then multiply them by 100 so that their units would be volatility of daily returns in percent.

the post 1997 period, however, unlike the result for shares, the difference is not statistically significant. This indicates that the decrease in the noise share in the more recent years of the sample is driven partly by a decrease in the total level of noise, but also by the large increases in variance attributable to information.

Table A.3 Panel C shows that smaller stocks tend to have a higher level of noise in their returns but also a higher level of private and public firm-specific information. There is no statistically significant difference in their level of market-wide information, although the point estimate suggests they have less variance driven by market-wide information. These results are largely in line with what we find for variance shares, except that in a relative sense (in shares) small stocks tend to have a lower proportion of the variance attributable to market-wide and public information.

Table A.3 Panel D shows the variance component levels by industry. The HiTech industry stands out as having one of the highest noise variances but also the highest information variance. This result is not so apparent in the variance shares reported in the paper because of the normalization by the total variance, which is higher in the HiTech industry.

Table A.3. Stock return variance components measured in levels.

This table reports means of the variance components measured in levels for the period from 1960 to 2015. Stock return variance is decomposed into market-wide information ($MktInfo$), private firm-specific information ($PrivateInfo$), public firm-specific information ($PublicInfo$), and noise ($Noise$). Each component is calculated separately as the standard deviation of percentage returns for each stock in each year, then averaged across stocks within the corresponding quartile or group, using a variance-weighted mean. Panel A reports full sample averages. Panel B splits the sample into two sub-periods from 1960 to 1996, and from 1997 to 2015. Panel C groups stocks into quartiles by size (market capitalization) with quartiles formed separately each year. Panel D groups stocks into major industry groups: the *Consumer* group comprises the industries Consumer Durables, NonDurables, Wholesale, Retail, and some Services (Laundries, Repair Shops); the $Healthcare$ group comprises the industries Healthcare, Medical Equipment, and Drugs; the *Manufact* group comprises the industries Manufacturing, Energy, and Utilities; the *HiTech* group comprises the industries Business Equipment, Telephone and Television Transmission; and the *Other* group comprises all other industries. We also report the differences in means for the post-1997 period minus the pre-1997 period (Panel B) and quartile 1 minus quartile 4 (Panel C) and corresponding t-statistics in parentheses. ***, **, and * indicate statistically significant differences at the 1%, 5%, and 10% levels using standard errors clustered by stock and by year. The sample consists of stocks listed on NYSE, AMEX, and NASDAQ.

	$MktInfo$	$PrivateInfo$	$PublicInfo$	$Noise$
Panel A: Full sample				
	1.75	3.28	4.34	3.98
Panel B: Sub-periods				
1960-1996	1.37	2.92	3.94	4.08
1997-2015	2.14	3.67	4.77	3.88
Difference (Post-Pre 1997)	0.77	0.75	0.83	-0.19
	(2.59)***	(3.84)***	(2.43)**	(-0.54)
Panel C: Quartiles by size (market capitalization)				
Q1=low	1.76	3.67	5.04	4.90
Q2	1.64	3.00	3.85	3.32
Q3	1.80	2.64	3.18	2.34
Q4=high	1.96	2.37	2.52	1.88
	-0.20	1.30	2.52	3.02
Difference (Q1-Q4)	(-0.97)	(5.96)***	(9.75)***	(12.00)***
Panel D: Industry groups				
Consumer	1.52	3.08	4.11	3.88
Healthcare	1.72	3.58	4.47	3.81
HiTech	2.02	3.65	4.68	4.06
Manufact	1.79	3.12	4.03	3.79
Other	1.64	3.08	4.32	4.14

3.2. Variance levels through time

Figure A.4 below is the analogue of Figure 5 in the paper, but for variance levels. It shows time-series trends in each of the variance components, controlling for compositional changes in the market by plotting the averages for different size groups.

The figure shows that all size groups have a similar trend with respect to the level of market-wide information (consistent with the variance shares), including the peaks during crises. Also consistent with the previous discussion of the results in Table A.3 above, smaller stocks have higher levels of noise, but also more variance attributable to information. The strongest trends in the variance components are for the firm-specific information, both public and private. The variance attributable to firm-specific information tends to increase through time for all size groups and this tendency is particularly strong for private firm-specific information. Finally, the level of noise shows a general tendency to decline since peaking in the late 1990s, although unlike the noise *shares*, the *level* of noise shows an additional tendency to spike during crises (such as 2008-2009) and episodes such as the dot com period (1999-2001).

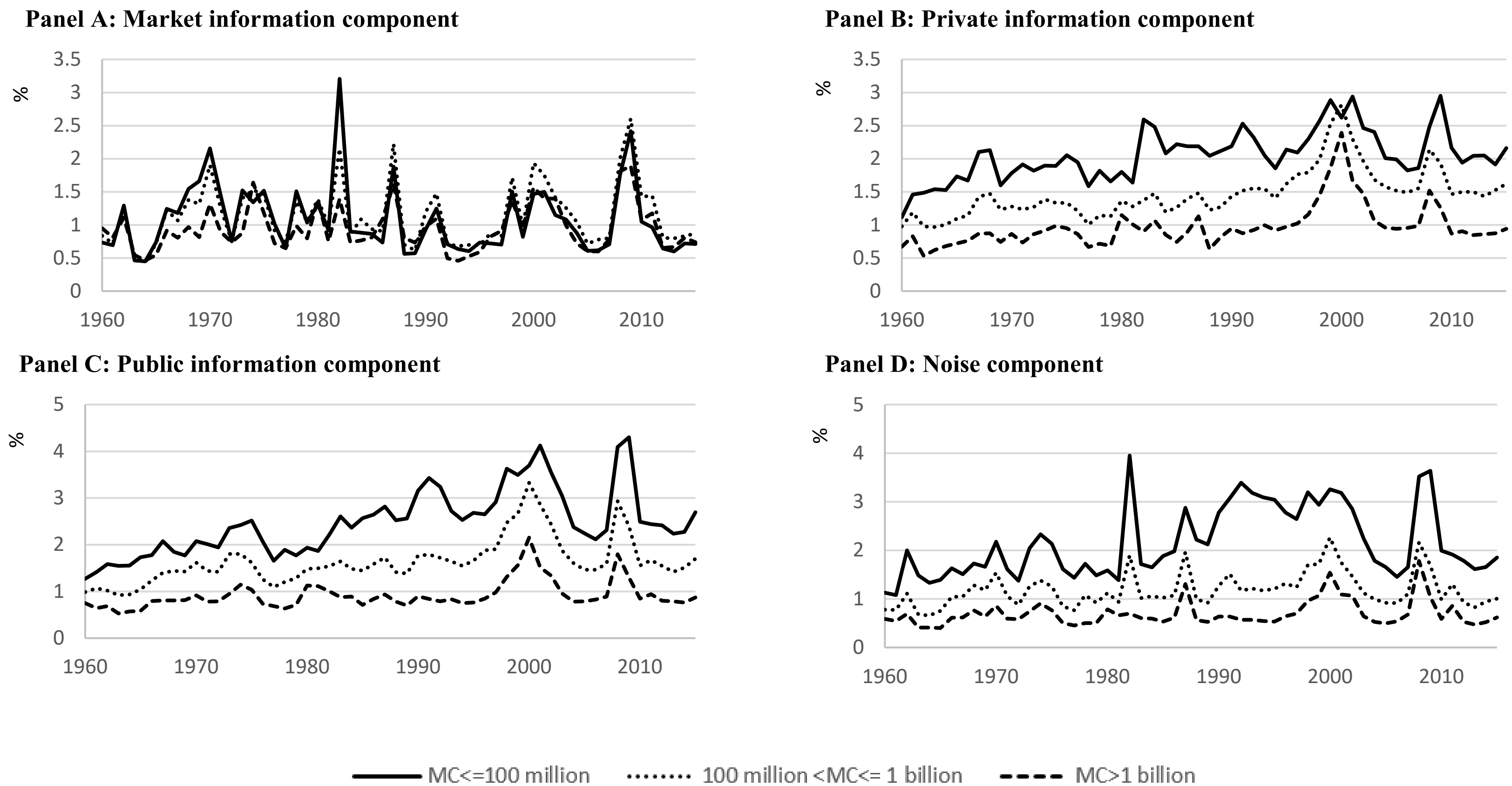


Figure A.4. Variance components measured in levels, reported by size groups through time.

This figure shows the time-series trends in stock return variance attributable to market-wide information (Panel A), private firm-specific information (Panel B), public firm-specific information (Panel C), and noise (Panel D) in three market capitalization groups: stocks with market capitalization less than $100 million, market capitalization between $100 million and $1 billion, and market capitalization greater than $1 billion. These breakpoints are in 2010 dollars and are adjusted for inflation forward and backward in time using the GDP price deflator. Each year stocks are assigned to one of the three groups based on their market capitalization at the start of the year. The variance components are calculated separately for each stock in each year (expressed as standard deviations of percentage returns) and then averaged for each size group in each year, using a simple-weighted mean. The sample consists of stocks listed on NYSE, AMEX, and NASDAQ from 1960 to 2015.

3.3. Variance levels and stock characteristics

Table A.4 below is the analogue of Table 3 in the paper but using variance levels rather than variance shares as the dependent variables. It reports the results of panel regressions in which the dependent variables are the variance components (in levels) from our variance decomposition and the independent variables include a range of stock characteristics and existing measures of information in prices as well as various control variables.

The results show that most of the relations documented in the paper for the variance shares are also found in variance levels. Below we summarize the results and point out the differences compared to using variance shares.

All three information components of variance are higher in the post 1997 period, controlling for other factors. This result suggests that prices have become more informative, even after accounting for changes in the composition of the market (industries, size), changes in institutional holdings, leverage, and liquidity. In contrast, the change in the total level of noise post 1997 is not statistically different from zero, controlling for other factors. That is not to say that the level of noise has not declined through time (it has), but rather, that the decline in noise is adequately explained by the other factors in the regressions such as increasing size and liquidity of companies, increasing institutional holdings, and so on.

Larger stocks tend to have less noisy prices (Panel A) and that tendency becomes statistically insignificant once we control for the fact that larger stocks also tend to have high institutional holdings and liquidity (Panel B). After accounting for those factors, larger stocks also tend to have more informative returns (Panel B).

The intraday/overnight variance ratio measure of information (French and Roll, 1986; Chordia et al., 2011) is positively related to all three information components of variance. Furthermore, the Hou and Moskowitz (2005) delay metric is associated with a higher level of noise in returns and less market-wide information, consistent with the results in the paper for variance shares. These results support the notion that the variance decomposition effectively identifies the different types of information in prices and separates them from the noise.

Hard-to-value stocks tend to have more volatile returns. The regressions suggest that the higher level of volatility is driven both by information (all three information-related variance components are higher for hard-to-value stocks) and noise (the level of noise also tends to be higher for such stocks). These findings are consistent with Kumar (2009) in that such stocks' individual investors display stronger behavioral biases (adding to noise), but also such stocks have higher levels of informed trading and are more sensitive to new information (adding to information variance). The difference from what we report in the paper using variance shares is that the noise share tends to be lower for such stocks whereas the noise level tends to be higher (as shown here), suggesting that the increased variance from information in such stocks is a stronger effect than the increased level of noise.

The proxies for limits to arbitrage (the two liquidity measures) are positively related to the total amount of noise in prices, consistent with our findings in the paper using variance shares. One of the liquidity measures (Amihud's ILLIQ) is also negatively related to the amount of private firm-specific information in prices and the amount of market-wide information in prices, consistent with illiquidity being a deterrent to information acquisition and informed trade. In contrast, both illiquidity measures are positively related with the amount of public firm-specific information in prices, consistent with studies that show firms tend to increase information disclosure in response to poor secondary market liquidity (e.g., Balakrishnan et al., 2014).

Leverage has a similar association with the informational components of variance as reported in the paper using variance shares. It has a positive association with public firm-specific information consistent with more disclosure by highly leveraged companies and a negative association with private firm-specific information consistent with a crowding out effect, although the relations are not as statistically significant as they are for variance shares.

Finally, companies with higher levels of institutional holdings tend to have less noisy prices, consistent with our findings in the paper for noise *shares* and consistent with other studies (e.g., Boehmer and Kelley, 2009; Sias and Starks, 1997). Institutional holdings are also associated with a lower variance of public and private firm-specific information. While this negative relation contrasts with our findings for variance shares, it is not statistically significant if we omit the hard-to-value proxy, which is related to institutional holdings.

Table A.4. Determinants of stock return variance levels.

This table reports the results from panel regressions of stock-year observations in which the dependent variables are components of stock return variance (measured as standard deviations of percentage returns) attributable to market-wide information ($MktInfo_{i,t}$), private firm-specific information ($PrivateInfo_{i,t}$), public firm-specific information ($PublicInfo_{i,t}$), and noise ($Noise_{i,t}$). The explanatory variables in Panel A are as follows. D_t^{POST} is a dummy variable that takes the value of one after 1997 and zero before. $lnP_{i,t}$ is the log price and $lnMC_{i,t}$ is the log market capitalization. $D_i^{Consumer}$, $D_i^{Healthcare}$, D_i^{HiTech}, and $D_i^{Manufact}$ are dummy variables that indicate the firm's industry group (the Other Industry grouping is the omitted category). $VarianceRatio_{i,t}$ is a measure of the amount of information impounded in prices during the trading session (volatility of open-to-close returns divided by the volatility of overnight close-to-open returns). $Delay_{i,t}$ is a measure of the delay with which a stock's prices respond to market-wide information. In Panel B we add an extended set of stock characteristics including the following. $InstoHold_{i,t}$ is the percentage of outstanding shares held by institutional investors. $HardToValue_{i,t}$ is a proxy for hard to value stocks, estimated as the first principal component of share turnover, idiosyncratic volatility, and firm age (following Kumar, 2009). $ILLIQ_{i,t}$ is the Amihud (2002) measure of illiquidity. $BidAskSpread_{i,t}$ is the average effective bid-ask spread. $Leverage_{i,t}$ is (long-term debt + current liabilities)/(long-term debt + current liabilities + shareholder's equity). The sample includes stocks listed on NYSE, AMEX, and NASDAQ from 1960 to 2015. T-statistics are in parentheses using standard errors clustered by stock and year. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels.

Panel A: Basic stock characteristics as determinants

Variable	$MktInfo_{i,t}$	$PrivateInfo_{i,t}$	$PublicInfo_{i,t}$	$Noise_{i,t}$
Intercept	1.86	2.38	3.59	3.53
	(7.63)***	(10.26)***	(12.74)***	(11.89)***
D_t^{POST}	0.25	0.19	0.34	-0.075
	(2.83)***	(2.21)**	(2.82)***	(-0.73)
$lnP_{i,t}$	-0.26	-0.59	-0.76	-0.64
	(-4.08)***	(-12.20)***	(-9.52)***	(-8.70)***
$lnMC_{i,t}$	-0.018	0.039	-0.062	-0.084
	(-1.00)	(2.91)***	(-3.80)***	(-4.66)***
$D_i^{Consumer}$	0.042	0.26	0.20	0.063
	(1.73)*	(10.18)***	(6.42)***	(1.87)*
$D_i^{Healthcare}$	0.17	0.62	0.47	0.11
	(3.13)***	(11.73)***	(8.23)***	(2.11)**
D_i^{HiTech}	0.41	0.70	0.64	0.29
	(3.23)***	(6.66)***	(4.59)***	(2.57)**
$D_i^{Manufact}$	0.043	0.13	0.071	0.006
	(1.28)	(4.27)***	(2.19)**	(0.21)
$VarianceRatio_{i,t}$	0.051	0.078	0.089	0.002
	(6.82)***	(6.73)***	(7.81)***	(0.16)
$Delay_{i,t}$	-0.92	0.34	0.40	0.60
	(-6.03)***	(2.20)**	(2.80)***	(4.81)***
R^2	20.03%	28.03%	36.76%	42.13%

Panel B: Extended set of stock characteristics as determinants				
Variable	$MktInfo_{i,t}$	$PrivateInfo_{i,t}$	$PublicInfo_{i,t}$	$Noise_{i,t}$
Intercept	1.66	1.80	2.61	2.27
	(6.82)***	(7.32)***	(9.17)***	(7.30)***
D_t^{POST}	0.24	0.18	0.39	0.084
	(2.97)***	(2.37)**	(3.96)***	(0.89)
$lnP_{i,t}$	-0.23	-0.43	-0.54	-0.36
	(-3.25)***	(-10.27)***	(-7.24)***	(-5.83)***
$lnMC_{i,t}$	0.023	0.082	0.013	-0.018
	(1.78)*	(3.86)***	(0.62)	(-1.04)
$D_i^{Consumer}$	0.11	0.37	0.29	0.12
	(2.79)***	(10.21)***	(7.30)***	(2.96)***
$D_i^{Healthcare}$	0.15	0.60	0.46	0.13
	(2.23)**	(12.33)***	(8.09)***	(2.51)**
D_i^{HiTech}	0.35	0.62	0.52	0.20
	(3.00)***	(7.98)***	(4.82)***	(2.37)**
$D_i^{Manufact}$	0.20	0.36	0.32	0.15
	(5.03)***	(10.33)***	(7.54)***	(3.93)***
$VarianceRatio_{i,t}$	0.042	0.066	0.083	0.005
	(4.98)***	(6.96)***	(8.75)***	(0.52)
$Delay_{i,t}$	-1.05	0.16	0.22	0.20
	(-6.81)***	(1.34)	(2.54)**	(2.43)**
$InstoHold_{i,t}$	-0.003	-0.003	-0.005	-0.004
	(-1.61)	(-3.17)***	(-3.03)***	(-2.57)**
$HardToValue_{i,t}$	0.30	0.49	0.51	0.26
	(5.89)***	(9.83)***	(7.69)***	(5.05)***
$ILLIQ_{i,t}$	0.00	-0.001	0.001	0.003
	(-1.89)*	(-3.54)***	(2.27)**	(3.16)***
$BidAskSpread_{i,t}$	0.001	0.003	0.007	0.012
	(0.87)	(1.62)	(2.34)**	(3.25)***
$Leverage_{i,t}$	-0.001	-0.003	0.004	-0.001
	(-0.64)	(-1.81)*	(1.64)	(-0.24)
R^2	29.02%	36.45%	43.84%	50.67%

4. Robustness of the VAR to different numbers of lags

The baseline VAR model in the paper uses five lags. The reasoning is that five lags are likely to capture most of the dynamics in response to shocks to trading, news, returns and so forth, yet keep the models simple and consistent across all stocks and years.

However, there is evidence that stocks sometimes take longer to fully digest information and respond to shocks. For example, Boguth, Carlson, Fisher, and Simutin (2016) show that for some stocks it can take up to 20 or more days to fully incorporate systematic information. We therefore examine alternative specifications for the VAR models, including using an extended set of lags, and analyzing how these modelling choices affect the variance decomposition results.

We start by testing for each stock in each year what is the optimal number of lags to include in the VAR according to the corrected Akaike Information Criterion (AIC). The optimal number of lags is the number that minimizes the AIC for that stock-year. Table A.5 below reports the distribution of optimal lag lengths for the stock-years in our sample.

The results show that in the pooled sample, for a typical (median) stock-year, two lags is optimal in the VAR, but in some stock-years considerably more lags are needed to adequately capture the dynamics. For example, in 5% of stock years (the 95th percentile) the optimal number of lags is at least 15. For the majority of our sample (at least 75% of the stock-years), five lags is sufficient as the 75th percentile of the optimal lag distribution is four lags.

Partitioning the sample into size quartiles each year, we see that smaller stocks tend to require more lags in the VAR, consistent with small stocks taking longer to fully reflect information and digest shocks. For example, the median stock-year in the small stock quartile has an optimal lag length of four and in 5% of the stock-years in the small stock quartile, the optimal lag length is at least 22. Therefore, a VAR model with five lags is adequate for most stock-years but there are some instances where more lags would produce a better fitting model.

Table A.5. Distribution of optimal lag length in stock-year observations
The table reports the percentiles of the distribution of optimal lag lengths in the VAR model specified in the paper, estimated separately each stock-year from daily data. The optimal lag length for each stock-year is given by the AIC.

	Percentiles of the distribution of optimal lag length				
	5^{th}	25^{th}	50^{th}	75^{th}	95^{th}
Pooled sample	1	1	2	4	15
Q1 = large stocks	1	1	2	3	10
Q2	1	1	2	3	11
Q3	1	1	2	5	17
Q4 = small stocks	1	2	4	8	22

Therefore, as the next step, we examine the robustness of the variance decomposition results to using a larger number of lags in those cases where it is warranted. Specifically, we use five lags for stock-years that have an optimal lag length of five or less according to AIC and double the number of lags to ten for all other stock-years.

Table A.6 below reports the average variance component shares using our baseline approach (five lags) and the extended number of lags. The overall estimates are very close for both approaches. The variance shares between the two approaches are within 1-2%, suggesting that the simpler approach of using five lags in the VARs for all stock-years provides a reasonable approximation.

We also check whether there is any difference in the dynamics of the variance components calculated from VAR models with an extended set of lags. Figure A.5 replicates Figure 3 of the paper but with the extended set of lags in the VAR model. Overall, the dynamics of the variance shares are also very similar under the two approaches.

Table A.6. Variance components estimated with different numbers of lags in the VAR.
Panel A reports the average variance shares in our baseline model using five lags in the VAR for all stock-days, as in the paper. Those results are the same as in Table 2 of the paper and are included for comparison. Panel B reports the average variance shares but doubling the number of lags in the VAR where it is warranted: for stock-years that have an optimal lag length of five or less according to AIC we use five lags and for all other stock-years we use ten lags. The stock return variance components are market-wide information ($MktInfoShare$), private firm-specific information ($PrivateInfoShare$), public firm-specific information ($PublicInfoShare$), and noise ($NoiseShare$). The sample is from 1960 to 2015 and consists of stocks listed on NYSE, AMEX, and NASDAQ.

	$MktInfoShare$ (%)	$PrivateInfoShare$ (%)	$PublicInfoShare$ (%)	$NoiseShare$ (%)
Panel A: Five lags (baseline results)				
	8.62	24.00	36.67	30.71
Panel B: Extended number of lags				
	8.69	23.63	35.71	31.97

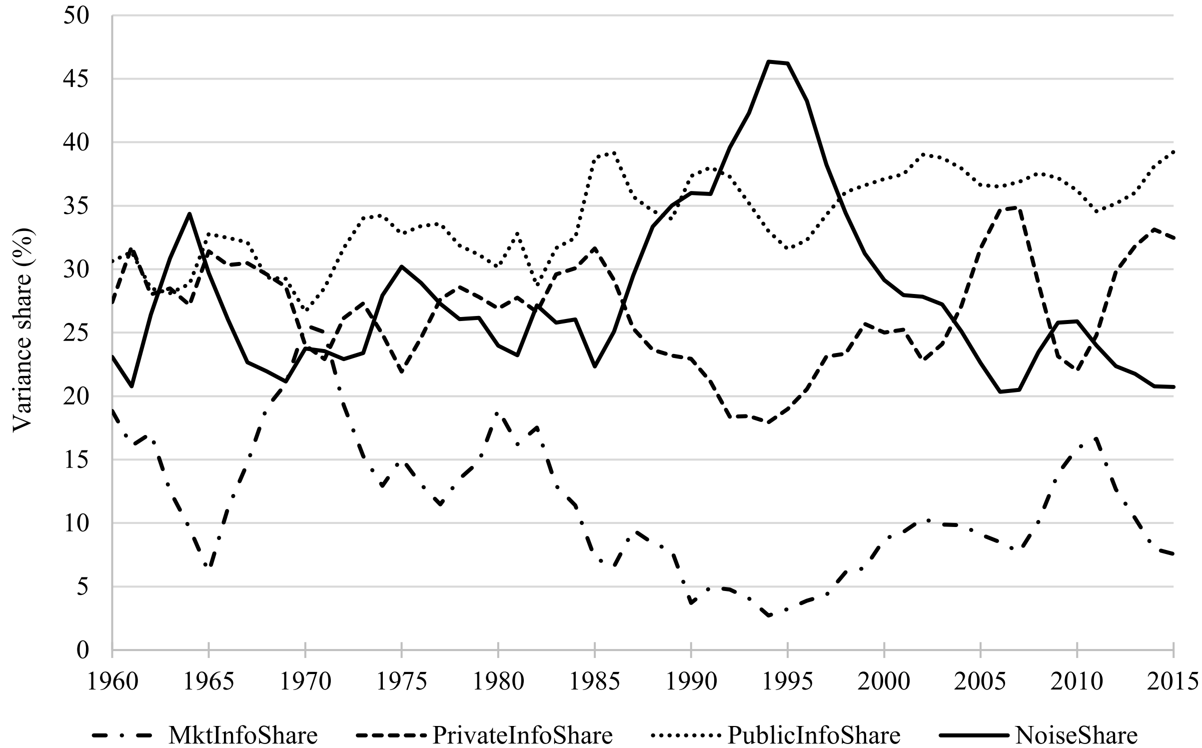


Figure A.5. Stock return variance components through time using an extended set of lags in the VAR.
This figure shows the time-series trends in the percentage of stock price variance that is attributable to noise ($NoiseShare$), market-wide information ($MktInfoShare$), trading on private firm-specific information ($PrivateInfoShare$), and public firm-specific information ($PublicInfoShare$). The variance shares are calculated separately for each stock in each year using VAR models that have five lags when the optimal lag length according to AIC is five or less, and ten lags otherwise. For each variance shares in each year we then take a variance-weighted average across stocks. The sample consists of stocks listed on NYSE, AMEX, and NASDAQ from 1960 to 2015.

5. Alternative measures of signed trading volume

To identify the private information that is impounded into prices through trading, we need a daily measure of buying activity minus selling activity. In the baseline variance decomposition in the paper, we use a simple proxy for signed dollar trading volume (signed dollar volume means buying minus selling dollar volume) similar to Pastor and Stambaugh (2003). The proxy is the dollar volume traded in the stock that day with the sign (plus or minus) applied from the sign of that day's return for the stock. The logic is that when there is excess buying, prices tend to move up and vice versa for excess selling, so the sign of the return approximates the direction of the imbalance between buyers and sellers, while the total dollar volume of trading gives a magnitude to the amount of trading activity.

The reason for using this simple proxy in the baseline model is twofold. First, it has very minimal data requirements: just daily volumes and returns. This allows the variance decomposition to be applied back to the 1960s or even before that and across many countries. The intraday data required for other proxies of signed dollar trading volume are only available in more recent time periods and in a limited number of countries and are far more computationally burdensome. Second, the measure is in the spirit of "bulk volume classification" (see Easley, Lopez de Prado, and O'Hara, 2016). By construction, it captures the price impact of trading, irrespective of whether the impact is from market orders, limit orders (the trade initiator or the passive side of trades) or both. It therefore is potentially better able to capture the private information that is in both order types and account for the fact that the passive side of a trade can also carry private information and impact prices, whereas not all other proxies for order imbalance have that property. We return to this point and elaborate on this issue later.

However, there are other measures of order imbalance or signed dollar volume that are widely used in the literature including measures that use more granular intraday data. One common approach is to sign individual trades using the Lee and Ready (1991) algorithm and then sum the dollar volume of buyer-initiated trades and subtract from it the sum the dollar volume of seller-initiated trades.

While this "trade initiator" measure of order imbalance has been used extensively, it suffers from some important drawbacks. The main drawback, illustrated by Easley, Lopez de Prado, and O'Hara (2016), is that by construction, trade-initiator based measures of order imbalance only capture the price impact and private information of the traders that trigger or initiate trades with marketable orders. A trade has two sides—one trader posts a bid or an offer quote (the "passive" side of the trade) and a second trader accepts that offer to trade by using a marketable order and thereby "triggers" or "initiates" the trade (the "active" side of the trade). Trade-initiator based measures of order imbalance look at each trade from the perspective of the trader that triggers the trade, neglecting the fact that the trader that posted the bid or ask quote in the first place may have private information that becomes reflected in prices by posting the bid or offer in the first place.

In other words, trade-initiator based measures of order imbalance are appropriate in scenarios in which bid and ask quotes are set by a market participant with little or no private information and informed traders are almost always trade initiators. While this scenario may have been a reasonable approximation in some pure dealer markets[2] (such as some equity markets in the 1980s), it is far from reality in limit order book markets such as US equity markets since at least the 1990s.[3] For example, O'Hara (2015) points out that large institutional investors typically act as trade initiators half the time and are on the passive side the other half of the time, suggesting that the "trade initiator" is a poor proxy for the activity of intrinsic buyers

[2] Even in US equity markets in the period in which designated market makers or "specialists" played a dominant role in liquidity provision, it is likely that some private information was impounded through the passive side of trades because designated market makers (that were often on the passive side) had informational privileges (such as seeing more details of order flow and the limit order book than other market participants) allowing them to incorporate private information in their quoted. Therefore, even in this era, the information in order imbalance measures is likely to capture only some fraction of the total private information that gets reflected in prices through trading.

[3] In the 1980s and 1990s, US equity markets such as the NYSE were "hybrid markets" meaning they had limit order books through which investors could trade directly with one another and designated market makers or "specialists"/dealers. For example, starting in 1976, investor orders in NYSE were transmitted electronically to specialists' screens over the DOT or SuperDot system. Incoming marketable orders would often be matched with resting limit orders of other market participants (which had precedence under exchange rules) without the need for the specialist to take the passive side of the trade (in the mid-1990s, specialists participated in around 17–18% of trades; Madhavan and Sofianos, 1998). Therefore, investors with private information could be trade initiators or could be on the passive side of a trade. Orders that were not filled immediately would remain in the order book, which became progressively more public and transparent through time. For example, from 1991 specialists were required to share order book information with floor traders. In 2002 NYSE introduced OpenBook, which allows traders off the NYSE floor to observe the depth in the book in real time at each price level. Boehmer, Saar, and Yu (2005) show that this increase in transparency further diminished the role of the specialists, allowing limit orders to be more actively managed and contribute to price discovery. Therefore, even as far back as the 1980s-1990s, measuring private information purely from the perspective of the trade initiator is likely to capture only some fraction of the private information that is impounded in prices and this fraction is likely to have diminished through time along with the evolution of equity markets.

and sellers in today's markets. In fact, the traders in today's markets that disproportionately act as the trade initiators tend to be the less informed retail traders that are less sophisticated in their approach to executing trades. Easley, Lopez de Prado, and O'Hara (2016) provide examples of how informed traders in today's markets impact market prices without ever being the trade initiator, for example, a "persistent bidder" that keeps posting orders at the bid but never "crossing the spread" to initiate a trade can push prices upwards as their private information gets impounded into prices. They show that trade-initiator based measures of order imbalance therefore do not adequately capture the information in trading. Consistent with these observations, Brogaard, Hendershott, and Riordan (2019) show that a lot of the price discovery in markets these days occurs through quote revisions, particularly by high-frequency traders, and Putnins and Michayluk (2018) show that informed traders are often on the passive side of trades and therefore limit orders convey private information to markets.

We therefore expect that trade-initiator based measures of order imbalance will capture only part of the private information in trading activity and thereby underestimate the amount of private information in prices.

In response to these deficiencies of the "trade initiator" Easley, Lopez de Prado, and O'Hara (2016) develop an alternative, "bulk volume classification" (BVC), to capture the information in trading activity accounting for the fact that the passive side of a trade can convey private information. Their approach is based on the premise that when there is buying pressure in markets, particularly when informed traders are buying (irrespective of their order types), prices tend to move up, and vice versa. Therefore, they use the information in returns (sign and magnitude) to assign an estimated imbalance between buyers and sellers over an interval of trading (hence "bulk", as opposed to signing individual trades). The estimated imbalance multiplied by the dollar volume in the interval gives the estimated signed dollar volume. Easley, Lopez de Prado, and O'Hara (2016) show in various tests that this approach of using the return to infer the imbalance between buyers and sellers is better able to capture the private information in trading activity than trade-initiator based measures of order imbalance.

The signed dollar volume proxy used in our baseline model (based on Pastor and Stambaugh, 2003) is similar in spirit to BVC, but simpler (drawing on less granular data and with less computational requirements). It too uses the return to infer the likely imbalance between buyers and sellers, without making any implicit assumptions about what order types informed buyers and sellers will use and whether the passive side of trades can contribute private information. It is therefore likely that the signed dollar volume proxy used in our baseline model will capture a larger fraction of the private information that is impounded into prices through trading. It is also likely to be more reliable through changes in market structure.[4]

To test these hypotheses, we re-estimate the VAR models and variance decomposition using trade-initiator based measures of order imbalance instead of the return-based proxy in our baseline model. To do this we must restrict our sample to 1993-2015 due to the more limited availability of intraday trade-level data, which we source from the TAQ database. For each trade during our sample period, we infer the "trade initiator" using the Lee and Ready (1991) algorithm. For each stock on each day, we sum the volume of buyer-initiated trades, subtract the volume of seller-initiated trades, then multiply this volume imbalance by the closing price to arrive at a signed dollar volume imbalance. That measure is then used as the x_t (signed dollar volume) variable in the variance decomposition. We also re-estimate our benchmark variance decomposition during this shorter sample period for comparison.

Table A.7 below reports the estimated average variance components using both measures of signed dollar volume. Consistent with our hypotheses based on the discussion above, the estimated fraction of variance that is attributed to private firm-specific information is much lower using the trade-initiator based measures of order imbalance (9.67%) than using the return-based measure of signed dollar volume

[4] For example, as equity markets transitioned from ones in which most informed traders were forced to be trade initiators because trading occurred predominantly through a dealer or market maker, to ones in which traders of all types interact directly with one another and informed traders are often on the passive side of trades, the trade-initiator based measures of order imbalance will go from capturing a large fraction of the private information to capturing a smaller fraction. This issue, which is less of a problem in return-based measures of order imbalance, can distort estimates of how the amount of private information in prices has changed through time (essentially the measurement error changes through time as equity markets evolve).

(24.17%). These results are consistent with the notion that the trade-initiator based measures of order imbalance underestimate the amount of private information that is impounded in prices through trading because of the substantial amount of price discovery that occurs through limit orders and the passive side of trades (e.g., O'Hara, 2015; Easley, Lopez de Prado, and O'Hara, 2016; Brogaard, Hendershott, and Riordan, 2019; Putnins and Michayluk, 2018).

What is perhaps even more interesting is how the other variance shares are affected by this misestimation. The noise share estimates are almost identical using the two measures and so too are the estimates of market-wide information. It is the estimated share of public firm-specific information that picks up the misestimation of the private firm-specific information. In other words, changing the measure of signed dollar volume does not affect inferences about the ratio of noise and information in prices. Nor does it affect the estimates split between firm-specific information and market-wide information. It merely affects the way the inferred firm-specific information is attributed between private information that enters through trading and public information. Mischaracterization of the direction and quantity of informed trading in the market leads to underestimation of how much of the firm-specific information in prices is impounded through trading and therefore overestimation of how much of the firm-specific information in prices is the result of public information.

We also examine the dynamics of the variance components estimated using the two measures of signed dollar volume. Figure A.6 below shows that the overall dynamics are similar, but the magnitudes of the components differ as observed previously. In both cases, the estimated share of private firm-specific information increases through time. The increases are approximately from 6% to 11% using the trade-initiator based measures of order imbalance and from 18% to 34% using the return-based measure of signed dollar volume. In both cases, the estimated share of private firm-specific information increases rapidly between 2003 and 2007—a period in which equity markets went through substantial automation of the trading process, leading to strong growth in algorithmic and high-frequency trading, large increases in turnover and trading volume accompanied by increasing liquidity and decreasing trading costs (e.g., Chordia, Roll, and Subrahmanyam, 2011).

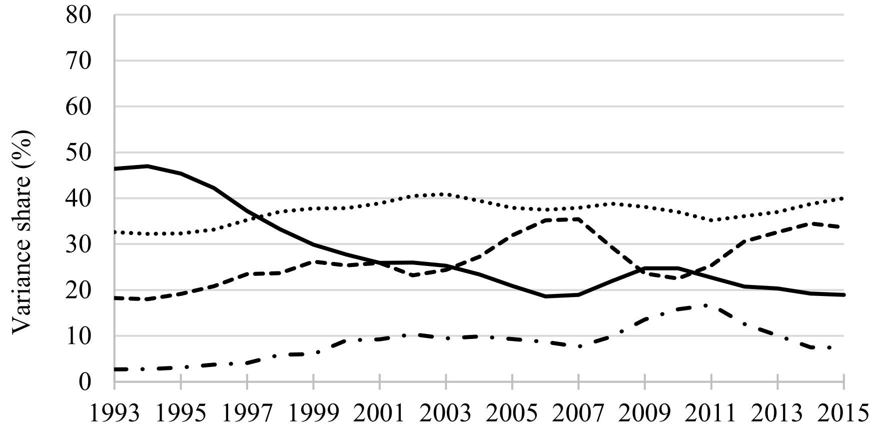
The amount by which the trade-initiator based measures of order imbalance underestimate private firm-specific information (as proxied by the gap between the two sets of estimates of this variance component) appears to increase through time (18%-6%=12% in 1993 vs 34%-11%=23% in 2015). This tendency is consistent with the notion that limit orders play an increasingly important role in price discovery in modern financial markets and therefore the trade-initiator becomes more limited in its ability to identify private information in markets (e.g., O'Hara, 2015; Easley, Lopez de Prado, and O'Hara, 2016).

The dynamics of the other variance components are approximately the same using the two different measures of signed dollar volume except the estimated level of public firm-specific information. Using the trade-initiator based measures of order imbalance, there appears to be much stronger growth in the amount of public firm-specific information in prices. However, as discussed earlier, because the private firm-specific information component is underestimated using trade-initiator based measures of order imbalance, the public firm-specific information in prices is overestimated leading to exaggerated estimates of its growth.

Table A.7. Stock return variance components using alternative measures of signed dollar volume.
This table reports the mean variance shares (expressed as percentages of variance) for the period from 1993 to 2015. Stock return variance is decomposed into market-wide information ($MktInfoShare$), private firm-specific information ($PrivateInfoShare$), public firm-specific information ($PublicInfoShare$), and noise ($NoiseShare$). In Panel A, the variance decomposition uses the Pastor and Stambaugh (2003) return-based measure of signed dollar volume. In Panel B, the variance decomposition uses the Lee and Ready (1991) trade initiator-based measure of signed dollar volume. The sample consists of stocks listed on NYSE, AMEX, and NASDAQ.

	$MktInfoShare$ (%)	$PrivateInfoShare$ (%)	$PublicInfoShare$ (%)	$NoiseShare$ (%)
Panel A: Baseline model using Pastor and Stambaugh (2003) return-based measure of signed dollar volume				
	8.55	24.17	37.22	30.05
Panel B: Alternative model using Lee and Ready (1991) trade initiator-based measure of signed dollar volume				
	8.42	9.67	51.55	30.37

Panel A: Baseline model using Pastor and Stambaugh (2003) return-based measure of signed dollar volume



Panel B: Alternative model using Lee and Ready (1991) trade initiator-based measure of signed dollar volume

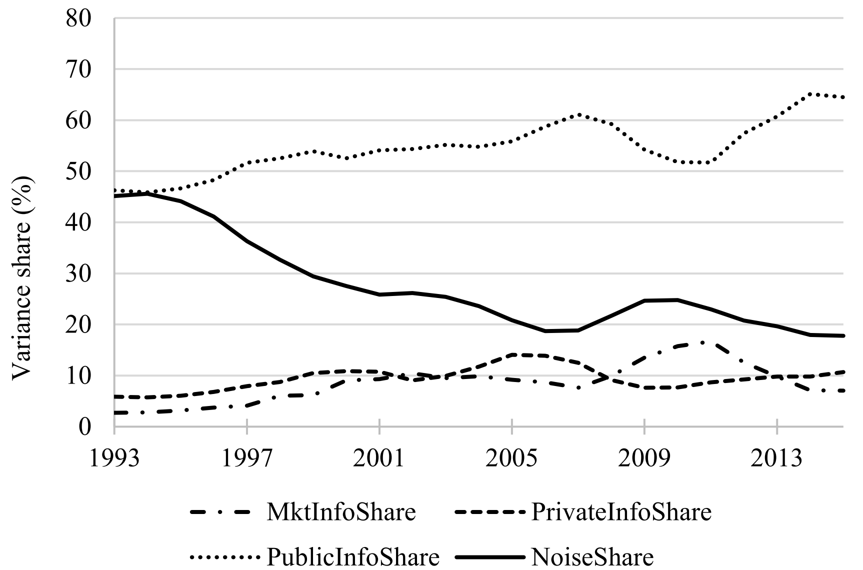


Figure A.6. Stock return variance components through time using different measures of signed dollar volume. This figure shows the time-series trends in the percentage of stock price variance that is attributable to noise ($NoiseShare$), market-wide information ($MktInfoShare$), trading on private firm-specific information ($PrivateInfoShare$), and public firm-specific information ($PublicInfoShare$). In Panel A, the variance decomposition uses the Pastor and Stambaugh (2003) return-based measure of signed dollar volume. In Panel B, the variance decomposition uses the Lee and Ready (1991) trade initiator-based measure of signed dollar volume. The variance shares are calculated separately for each stock in each year. Then for each variance shares in each year we then take a variance-weighted average across stocks. The sample consists of stocks listed on NYSE, AMEX, and NASDAQ from 1993 to 2015.

6. The impact of dealer collusion on noise in prices

Christie and Schultz (1994) find evidence of collusive behavior by NASDAQ dealers during a period from 1991 until the collusive behavior was exposed a few years later. They show that NASDAQ dealers colluded to maintain artificially wide spreads by avoiding odd-eighth quotes. This behavior increased the effective tick size, which could increase noise due to bid-ask bounce. While the times-series of noise shares in the paper (Figures 3 and 5) and noise levels in this Internet Appendix (Figure A.4) are consistent with the notion that collusion in the 1990s increased the level and share of noise in prices, we turn to more formal tests of this hypothesis.

A natural control group is non-NASDAQ stocks. We therefore take a subsample of four years before and during the collusive behavior and estimate a difference-in-differences model that uses non-NASDAQ stocks as a control group:

$$Noise_{it} = \alpha + \beta D_t^{COLLUSION} + \gamma D_t^{COLLUSION} NASDAQ_i + \delta NASDAQ_i + \varepsilon_{it},$$

where $D_t^{COLLUSION}$ is an indicator variable that takes the value of one in the collusion period (1991-1994) and zero otherwise and the indicator variable $NASDAQ_i$ is an indicator variable that takes the value one for NASDAQ-listed stocks and zero otherwise. We estimate the model for both the level of noise and the noise share.

Table A.8 below reports the results. The results show that during the period of collusion by NASDAQ dealers the returns of NASDAQ-listed stocks become significantly noisier, both in an absolute sense (levels) and a relative sense (shares), consistent with discreteness in price grids contributing to pricing errors and noise in returns. The magnitude of the effect is economically meaningful. The increase in the noise share for NASDAQ-listed stocks is estimated to be 9.26% of variance, which is large considering that the pooled sample mean noise share is around 30.71% of variance.

Table A.8. Effects of dealer collusion on noise.
This table reports the results from difference-in-differences regressions in which the dependent variable is the level (Model 1) and share (Model 2) of noise in prices estimated from the variance decomposition described in the paper. The models examine the impact of collusion by NASDAQ dealers to avoid odd-eighth quotes, using four years before and four years during the collusion (1987 to 1994). $D_t^{COLLUSION}$ takes the value of one in the collusion period (1991-1994) and zero before. $NASDAQ_i$ is one for NASDAQ-listed stocks and zero otherwise. t-statistics are in parentheses using standard errors clustered by stock. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels. The sample includes all stocks listed on NYSE, AMEX, and NASDAQ.

Variable	Model 1 $Noise_{i,t}$	Model 2 $NoiseShare_{i,t}$
Intercept	1.48	20.89
	(73.68)***	(111.17)***
$D_t^{COLLUSION}$	-0.12	-2.16
	(-6.17)***	(-10.73)***
$D_t^{COLLUSION} \times NASDAQ_i$	0.65	9.26
	(21.90)***	(26.01)***
$NASDAQ_i$	0.77	10.29
	(30.30)***	(31.08)***
R^2	11.7%	12.5%

7. Robustness to lower winsorization level

We re-estimate the variance decomposition model and rather than winsorizing the variance components at the 5% and 95% levels (percentiles), we instead winsorize at a lower level of 1% and 99%. Figure A.7 and Table A.9 are equivalent to Figure 3 and Table 2 in the paper but using these lower levels of winsorization. The results are very similar, suggesting that they are not overly sensitive to the choice of the two winsorization levels.

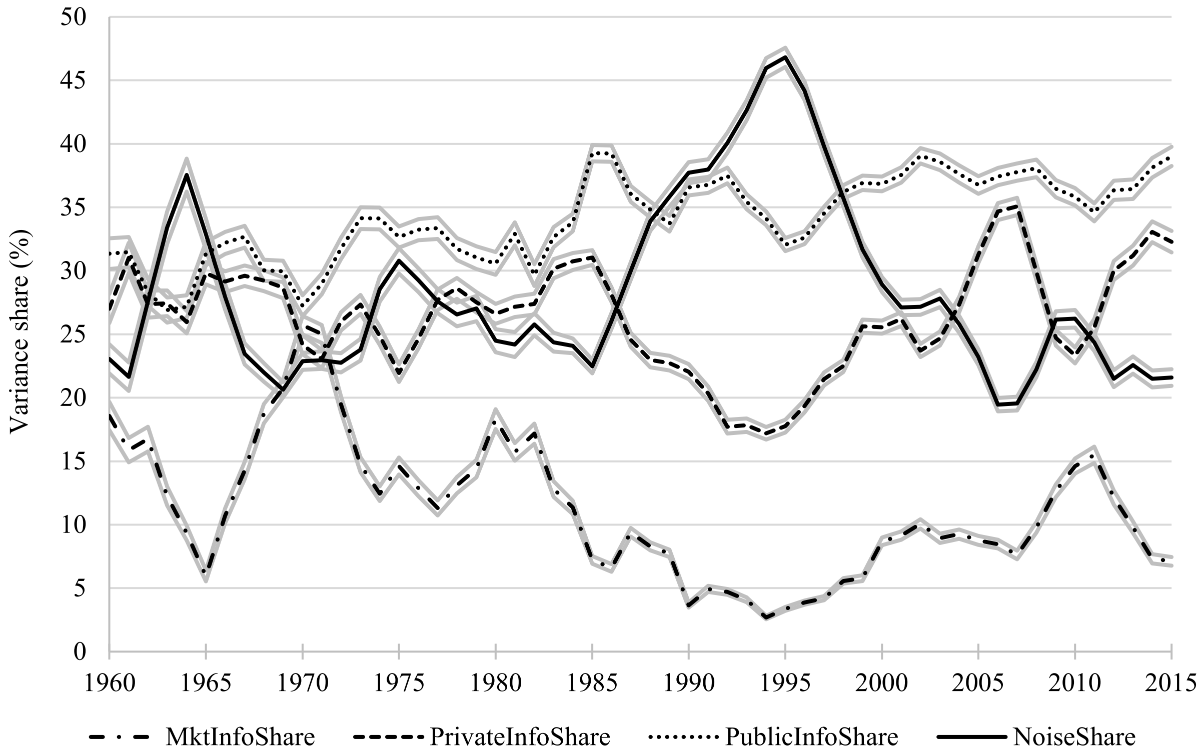


Figure A.7. Stock return variance components for US stocks through time using a lower level of winsorization.
This figure shows the time-series trends in the percentage of stock price variance that is attributable to noise ($NoiseShare$), market-wide information ($MktInfoShare$), trading on private firm-specific information ($PrivateInfoShare$), and public firm-specific information ($PublicInfoShare$). The variance shares are calculated separately for each stock in each year based on a VAR model. For each variance shares in each year we then take an average across stocks. Light gray lines provide 99% confidence intervals. The sample consists of stocks listed on NYSE, AMEX, and NASDAQ from 1960 to 2015. In contrast to Figure 3 of the paper, here we winsorize the variance components at the 1% and 99% levels.

Table A.9. Stock return variance components in the baseline model using a lower level of winsorization.

This table reports the mean variance shares (expressed as percentages of variance) for the period from 1960 to 2015. Stock return variance is decomposed into market-wide information ($MktInfoShare$), private firm-specific information ($PrivateInfoShare$), public firm-specific information ($PublicInfoShare$), and noise ($NoiseShare$). Panel A reports full sample averages. Panel B splits the sample into two sub-periods from 1960 to 1996, and from 1997 to 2015. Panel C groups stocks into quartiles by size (market capitalization) with quartiles formed separately each year. Panel D groups stocks into major industry groups: the *Consumer* group comprises the industries Consumer Durables, NonDurables, Wholesale, Retail, and some Services (Laundries, Repair Shops); the *Healthcare* group comprises the industries Healthcare, Medical Equipment, and Drugs; the *Manufact* group comprises the industries Manufacturing, Energy, and Utilities; the *HiTech* group comprises the industries Business Equipment, Telephone and Television Transmission; and the *Other* group comprises all other industries. The variance component shares are calculated separately for each stock in each year then averaged across stocks within the corresponding quartile or group, using the variance-weighted mean. The 99% confidence intervals are in square brackets. We also report the differences in means for the post-1997 period minus the pre-1997 period (Panel B) and quartile 1 minus quartile 4 (Panel C) and corresponding t-statistics in parentheses. ***, **, and * indicate statistically significant differences at the 1%, 5%, and 10% levels using standard errors clustered by stock and by year. The sample consists of stocks listed on NYSE, AMEX, and NASDAQ (average of 4,362 stocks per year with a total of 22,025 stocks). In contrast to Table 2 of the paper, here we winsorize the variance components at the 1% and 99% levels.

	$MktInfoShare$ (%)	$PrivateInfoShare$ (%)	$PublicInfoShare$ (%)	$NoiseShare$ (%)
Panel A: Full sample				
	8.34	23.55	35.75	32.36
	[8.28-8.40]	[23.46-23.64]	[35.65-35.85]	[32.24-32.48]
Panel B: Sub-periods				
1960-1996	7.11	21.66	34.23	36.99
	[7.03-7.19]	[21.54-21.78]	[34.11-34.36]	[36.83-37.16]
1997-2015	9.63	25.53	37.34	27.50
	[9.53-9.72]	[25.39-25.68]	[37.19-37.49]	[27.34-27.66]
Difference (Post-Pre 1997)	2.52	3.87	3.11	-9.49
	(1.65)*	(2.59)**	(3.21)***	(-4.85)***
Panel C: Quartiles by size (market capitalization)				
Q1=low	4.87	21.03	35.61	38.49
	[4.80-4.94]	[20.86-21.21]	[35.42-35.80]	[38.25-38.72]
Q2	8.84	25.04	37.14	28.98
	[8.72-8.95]	[24.86-25.23]	[36.93-37.35]	[28.75-29.21]
Q3	14.74	28.00	36.59	20.67
	[14.59-14.90]	[27.81-28.19]	[36.39-36.78]	[20.50-20.85]
Q4=high	21.83	30.36	31.22	16.60
	[21.65-22.01]	[30.17-30.54]	[31.04-31.40]	[16.47-16.72]
Difference (Q1-Q4)	-16.96	-9.32	4.39	21.89
	(-15.82)***	(-5.86)***	(5.12)***	(13.41)***
Panel D: Industry groups				
Consumer	7.74	22.90	35.55	33.82
	[7.61-7.87]	[22.70-23.10]	[35.33-35.76]	[33.55-34.08]
Healthcare	7.34	27.68	37.14	27.84
	[7.16-7.53]	[27.33-28.04]	[36.77-37.50]	[27.44-28.23]
HiTech	9.41	25.30	36.27	29.03
	[9.26-9.55]	[25.08-25.51]	[36.04-36.50]	[28.77-29.29]
Manufact	9.50	23.83	34.03	32.64
	[9.36-9.64]	[23.65-24.02]	[33.83-34.22]	[32.39-32.89]
Other	7.45	20.86	35.93	35.76
	[7.34-7.56]	[20.69-21.02]	[35.76-36.11]	[35.54-35.98]

8. Motivating theoretical model

8.1. Model

The model starts with the firm's cash flow generating process:

$$C_t = K_0 X_t, \tag{A.1}$$

where C_t is the future cash flow at time t, K_0 is the initial investment, and X_t captures the random shocks to the cash flow process. Departing from the Jin and Myers (2006) model, in which the focus is solely on information innovations, we suppose that X_t, as it is perceived or estimated by investors ($\widetilde{X_t}$), is driven by four components, namely three information-driven components and noise:

$$\widetilde{X_t} = \theta_{1,t} + \theta_{2,t} + \theta_{3,t} + \theta_{4,t}. \tag{A.2}$$

The term $\theta_{1,t}$ reflects market-wide cash flow information, while $\theta_{2,t}$ and $\theta_{3,t}$ reflect firm-specific cash flow information, and $\theta_{4,t}$ reflects noise. The distinction between $\theta_{2,t}$ and $\theta_{3,t}$ is that $\theta_{2,t}$ is firm-specific information revealed through trading on private information, while $\theta_{3,t}$ is firm-specific information revealed through public information such as company announcements and news. Firm-specific information ($\theta_{2,t}$ and $\theta_{3,t}$) is orthogonal to market-wide information ($\theta_{1,t}$) and noise is assumed to be uncorrelated with information.

The noise ($\theta_{4,t}$) is a reduced form way to account for many sources of noise in prices. It accounts for the fact that prices set by investors following their beliefs about $\widetilde{X_t}$ can deviate from intrinsic values that are based solely on true information. The deviations can arise from rational causes such as imperfect signals, or various friction, or irrational reasons and biases. The addition of $\theta_{4,t}$ to the model is in the spirit of Black's (1986) notion that "noise trading is trading on noise as if it were information", i.e., investors believe $\theta_{4,t}$ to be information about cash flows, causing prices to depart from their efficient intrinsic values. The ultimate effect of injecting noise into Equation (A.2) is to make prices noisy, which can be interpreted as representing the effects of frictions such as a discrete pricing grid, non-synchronous trading and so on.

Similar to Jin and Myers (2006), we assume that all of the information and noise components follow stationary AR(1) processes driven by a set of random shocks ($\varepsilon_{1,t}$, $\varepsilon_{2,t}$, $\varepsilon_{3,t}$, and $\varepsilon_{4,t}$).[5] That is, $\theta_{i,t+1} = \theta_{i,0} + \varphi\theta_{i,t} + \varepsilon_{i,t+1}$, where $0 < \varphi < 1$. We define r as the discount rate and K_t as the investors' valuation

[5] This and other assumptions made in this appendix are for the convenience of solving the motivating theoretical model (they mainly follow from Jin and Myers (2006)). We do not rely on these assumptions in the empirical model.

of the firm, which is the present value of future cash flows (assuming each period's cash flow is paid out), conditional on the information that the investors have at date t about $\widetilde{X_t}$:

$$K_t = PV\{E(C_{t+1}|I_t), E(C_{t+2}|I_t), \dots; r\}. \tag{A.3}$$

For investors $I_t = \{\theta_{1,t} + \theta_{2,t} + \theta_{3,t} + \theta_{4,t}\}$ and therefore the value of K_t is:

$$K_t = \frac{K_0 X_0}{r(1-\varphi)} - \frac{K_0 X_0 \varphi}{(1+r-\varphi)(1-\varphi)} + \frac{\varphi}{1+r-\varphi} K_0(\theta_{1,t} + \theta_{2,t} + \theta_{3,t} + \theta_{4,t}). \tag{A.4}$$

Let $\tilde{r}_{t+1}$ be the total realized return on the firm's shares in the period $t+1$. The return $\tilde{r}_{t+1}$ is calculated as the change in the investors' valuation of the firm from one period to the next plus that period's cash flow, which is paid out, and is therefore a function of the shocks to the investors' information about the cash flow process:

$$\tilde{r}_{t+1} = r + b_t\left(\varepsilon_{1,t+1} + \varepsilon_{2,t+1} + \varepsilon_{3,t+1} + \varepsilon_{4,t+1}\right), \tag{A.5}$$

where

$$b_t = \frac{(1+r)}{\frac{X_0(1+r)}{r} + \varphi(\theta_{1,t} + \theta_{2,t} + \theta_{3,t} + \theta_{4,t})}. \tag{A.6}$$

The derivation of Equations (A.5) and (A.6) is provided below in Appendix A.2 (Proof 1). Equation (A.6) implies that the random component of realized stock returns, $b_t\left(\varepsilon_{1,t+1} + \varepsilon_{2,t+1} + \varepsilon_{3,t+1} + \varepsilon_{4,t+1}\right)$, is driven by shocks to the various types of information (market-wide information, firm-specific information revealed through trading on private information, and public firm-specific information) as well as innovations in noise.

The model suggests the stock return variance can be decomposed into four distinct sources of variation. Performing a variance decomposition on the realized returns in this theoretical framework serves as a guide for the empirical decomposition. Because of the independence between the components of realized returns, the variance of realized returns is equal to the sum of the contributions from each of the components. This allows us to define a set of variance shares as:

$$\eta_j = \frac{Var(\varepsilon_{j,t})}{Var(\varepsilon_{1,t} + \varepsilon_{2,t} + \varepsilon_{3,t} + \varepsilon_{4,t})}, \tag{A.7}$$

with subscript $j = \{1, 2, 3, 4\}$ denoting each return component. The first of these variance shares, η_1, is the contribution of market-wide information to an individual stock's variance. The second (η_2) is the contribution of firm-specific information that is revealed through trading (private firm-specific information), while the third (η_3) is the contribution of public firm-specific information. The last component (η_4) is the effect of noise on stock return variance.

8.2. Proofs

Proof 1.

The total realized return on the firm's shares for outsiders is calculated as the change in the investors' valuation of the firm from one period to the next including the cash flow that is paid out:

$$\tilde{r}_{t+1} = \frac{K_{t+1} + C_{t+1}}{K_t} - 1. \tag{A.8}$$

Substituting Equation (A.4) for K_t into (A.8) and rearranging gives:

$$\tilde{r}_{t+1} = \frac{\left(\frac{1+r}{1+r-\varphi}\right) C_{t+1} - \frac{\varphi}{1+r-\varphi} C_t}{\frac{(1+r)}{r} \frac{K_0 X_0}{(1+r-\varphi)} + \frac{\varphi}{1+r-\varphi} C_t}. \tag{A.9}$$

For investors $I_t = \{\theta_{1,t} + \theta_{2,t} + \theta_{3,t} + \theta_{4,t}\}$ and therefore,

$$\tilde{r}_{t+1} = \frac{\left(\frac{1+r}{1+r-\varphi}\right) K_0 \sum_{j=1}^{4} \theta_{j,t+1} - \frac{\varphi}{1+r-\varphi} K_0 \sum_{j=1}^{4} \theta_{j,t}}{\frac{(1+r)}{r} \frac{K_0 X_0}{(1+r-\varphi)} + \frac{\varphi}{1+r-\varphi} K_0 \sum_{j=1}^{4} \theta_{j,t}}. \tag{A.10}$$

Multiplying the denominator and the numerator by $(\frac{1+r-\varphi}{K_0})$ and rearranging, we get:

$$\tilde{r}_{t+1} = r + \frac{(1+r)\left[\sum_{j=1}^{4}\theta_{j,t+1} - \varphi\sum_{j=1}^{4}\theta_{j,t} - X_0\right]}{\frac{(1+r)X_0}{r} + \varphi\sum_{j=1}^{4}\theta_{j,t}}. \tag{A.11}$$

Given the assumptions that all of the information and noise components follow stationary AR(1) process (we do not make this assumption in the empirical model), Equation (A.11) becomes:

$$\tilde{r}_{t+1} = r + b_t\left(\varepsilon_{1,t+1} + \varepsilon_{2,t+1} + \varepsilon_{3,t+1} + \varepsilon_{4,t+1}\right), \tag{A.12}$$

where

$$b_t = \frac{(1+r)}{\frac{(1+r)X_0}{r} + \varphi(\theta_{1,t} + \theta_{2,t} + \theta_{3,t} + \theta_{4,t})}. \tag{A.13}$$

Proof 2.

The realized return on stock i in Equation (A.12) can be rewritten as:

$$\tilde{r}_{i,t+1} = r + b_t\varepsilon_{1,t+1} + b_t\varepsilon_{f,t+1}, \tag{A.14}$$

where

$$b_t = \frac{(1+r)}{\frac{(1+r)X_0}{r} + \varphi(\theta_{1,t} + \theta_{2,t} + \theta_{3,t} + \theta_{4,t})},$$

and $\varepsilon_{f,t+1}$ is the shock related to firm-specific information and noise,

$$\varepsilon_{f,t+1} = \varepsilon_{2,t+1} + \varepsilon_{3,t+1} + \varepsilon_{4,t+1}. \tag{A.15}$$

The market return is the same as the return of a stock with no idiosyncratic risk:

$$\tilde{r}_{m,t+1} = r + d_t\varepsilon_{1,t+1}, \tag{A.16}$$

where

$$d_t = \frac{(1+r)}{\frac{(1+r)X_0}{r} + \varphi\theta_{1,t}}. \tag{A.17}$$

From Equation (A.16), we have:

$$\varepsilon_{1,t+1} = \frac{r_{m,t+1} - r}{d_t}. \tag{A.18}$$

Substitute the expression in Equation (A.18) for $\varepsilon_{1,t+1}$ into (A.14):

$$\tilde{r}_{i,t+1} = r + \frac{b_t}{d_t} r_{m,t+1} - \frac{b_t}{d_t} r + b_t \varepsilon_{f,t+1}. \tag{A.19}$$

Conditional on $\theta_{1,t}$, $\theta_{2,t}$, $\theta_{3,t}$ and $\theta_{4,t}$, the stock return variance is therefore a function of shocks to the investors' information about the cash flow process:

$$Var(\tilde{r}_{i,t+1}) = \left(\frac{b_t}{d_t}\right)^2 Var(r_{m,t+1}) + Var(b_t \varepsilon_{f,t+1}) \tag{A.20}$$

Substituting (A.16) into Equation (A.20), and rearranging, we get:

$$Var(\tilde{r}_{i,t+1}) = b_t^2 \left[Var(\varepsilon_{1,t+1}) + Var(\varepsilon_{2,t+1}) + Var(\varepsilon_{3,t+1}) + Var(\varepsilon_{4,t+1})\right] \tag{A.21}$$

From Equation (A.19), the proportion of variance explained by the market, R^2, can be written as:

$$\begin{aligned} R^2 &= \frac{\left(\frac{b_t}{d_t}\right)^2 Var(r_{m,t+1})}{Var(\tilde{r}_{f,t+1})} \\ &= \frac{Var(\varepsilon_{1,t+1})}{Var(\varepsilon_{1,t+1}) + Var(\varepsilon_{2,t+1}) + Var(\varepsilon_{3,t+1}) + Var(\varepsilon_{4,t+1})} \\ &= \frac{1}{1 + \frac{1}{\eta_1}(\eta_2 + \eta_3 + \eta_4)}. \end{aligned} \tag{A.22}$$

9. Accounting for the covariance between noise and information

In computing the variance shares in Equation (10) of the paper, we ignore the covariance between information (innovations in the efficient price) and noise (changes in the pricing error). Here we show that accounting for this covariance has little effect on our estimates of the variance shares.

One way to account for the covariance term is to distribute it between the information components of variance and the noise component of variance in the same proportions as the variances of these components and then recompute the variance shares from the covariance-adjusted components using the total return variance as the normalizing variable. In this approach, we allocate a fraction α of $2cov(w_t, \Delta s_t)$ to the information variance and a fraction $(1-\alpha)$ to the noise variance, where $\alpha = \frac{\sigma_w^2}{\sigma_w^2+\sigma_s^2}$. Consequently, the information and noise shares of variance become:

$$\begin{aligned}
InfoShare &= \left(\sigma_w^2 + \frac{\sigma_w^2}{\sigma_w^2+\sigma_s^2} 2cov(w_t, \Delta s_t)\right) / \sigma_r^2 \\
&= \sigma_w^2 \left(1 + \frac{2cov(w_t, \Delta s_t)}{\sigma_w^2+\sigma_s^2}\right) / \sigma_r^2 = \sigma_w^2 \left(\frac{\sigma_w^2 + 2cov(w_t, \Delta s_t) + \sigma_s^2}{\sigma_w^2+\sigma_s^2}\right) / \sigma_r^2 \\
&= \sigma_w^2 \left(\frac{\sigma_r^2}{\sigma_w^2+\sigma_s^2}\right) / \sigma_r^2 = \sigma_w^2 / (\sigma_w^2+\sigma_s^2) \\
NoiseShare &= \left(\sigma_s^2 + \frac{\sigma_s^2}{\sigma_w^2+\sigma_s^2} 2cov(w_t, \Delta s_t)\right) / \sigma_r^2 \\
&= \sigma_s^2 \left(1 + \frac{2cov(w_t, \Delta s_t)}{\sigma_w^2+\sigma_s^2}\right) / \sigma_r^2 = \sigma_s^2 \left(\frac{\sigma_r^2}{\sigma_w^2+\sigma_s^2}\right) / \sigma_r^2 \\
&= \sigma_s^2 / (\sigma_w^2+\sigma_s^2)
\end{aligned} \tag{A.23}$$

Equation (A.23) shows that after the distribution of the covariance term into information and noise components, we have exactly the same variance shares as in the baseline (Equation (10) of the paper).

An alternative way of distributing the covariance term is to add it entirely to either the information component of variance or to the noise component of variance thereby producing upper and lower bounds on the variance shares. Applying this approach, we find that the upper and lower bounds are extremely narrow (e.g., the noise share has a lower bound of 25.73% and an upper bound of 27.86%, whereas the information share has a lower bound of 72.14% and an upper bound of 74.27%). Therefore, ignoring the covariance term in the baseline variance shares has little effect on the results.

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